priv-csrs.html

!"#$ % & '( )* % +, -
./

SYSTEM RISC-V ISA . !
" # . Zicsr $%& ' , ( ) *+ ,-./ (CSR) 0123 45 -'" -65 7 ,
8 9- . :;<=& Zicsr $% >? . @ A& B C
>? - :;<= 5D EF& GH IHJ .

KLM N O (I & N P *+ Q&R SC CSRT UVW " # . CSRX (I CSR
Y Z "[&O \]W " # . %X CSR ^ _` ab N . c %&O
"[d CSR 5De , Zf23 ' "[& g\ h ij @ A N .

CSRe C ' "[& kj #lHR , @m nX "[&OR \]W " # o&
p qr .

s[CSR& 45 " teC u-v , 65 " teC #T " # .

s[RISC-V ISA wx 4,096y CSRT zq 12P{ _` T $m . i|* CSR ^ *z
4P{ X s 1e } "[& G8 45) 65 \] CD Y . w*z 2P{
X ,-./C 45 /65 CD (, ,) - , 45 ~ ( ) - •€ .

• 2P{ X q, CSR& \]W " # w*f* "[T „ , P{ …† X ‡^ *f*
CSRT •€ .

!" #$% & ' (
)* +) ,- .)* /01 +2 3 456 7 8 93. : ; 9 <=> ?@
ABC 1 DEF= G 8 93 H IJKLM N 9 < OP 7 8 93. Q 1
RSET

‰Š - ‹ CSR& \] Œ•j # . 2Ž "[u CSR& \] • • ‘’
ŒQ “ ”{• 0 –& N C* ŒQ (virtual-instruction exception)C —˜ . 45 ~
,-./& 6• • ‘’ ŒQC —˜ . 45 /65 ,-./ ™&R Z 45 ~ P{C #T "
#3„ , š› q, 45 ~ P{& x 65 œ• .

s 1X s[Rž 1 ' R CSR ^ W, Ÿ R •€ . 1 ' R -' CSR
^ ¡c s[$%&O Š' - ‹ .

9¢ s[45 /65 CSR £¤@ •.¥ 3 Œ• . ¦ 9¢
&O \]W " #-v , £¤@ &Ov m . KLM 9¢ C c1 ,-./
EF& \] • ‘’ ŒQ —˜•; § ¨©

UVW : %" EF : % XP Y1F :Z [S\ :] ^_C& 8 : `a
bcd HC T 7 8 931F e fB G 8 931F e g :Z M -h Pi

j= 1 k= l,md" n oM p ABC 7 8 93. 1 H IJKL, qrpF
st 1 H IJKu 9 < 5v wxd- y k= z

01 +, 23

01	01	01	45 +,	"& 67 89:
;< !"= 5>				
				> ? @
				> ? @
				!"= A ? @
				> ? B"
				> ? B"
				!"= A ? B"
CDE F 5>				
				> ? @
				> ? @
				> ? @
				!"= A ? @
				> ? @
				> ? @
				!"= A ? @
				> ? B"
				> ? B"
				!"= A ? B"
# DE F				
				> ? @
				> ? @
				> ? @
				!"= A ? @
				> ? @
				> ? @
				!"= A ? @
				> ? B"
				> ? B"
				!"= A ? B"
GH 5>				
				> ? @

01	01	01	45 +,	"& 67 89:
				> ? @
				> ? @
				> ? @ IJK
				IJK L* B"
				!"= A ? @
				> ? @
				> ? @
				!"= A ? @
				> ? B"
				> ? B"
				!"= A ? B"

s 2~s 5 LŠ CSR ^C W, CSRT ª . • 9 , «¬/) -^"o CSRX s[P CSR . 9- ,-./ c %& N x &O . KLM& ,-./C >? §X :®H o& p qr .

/. 23M ;< 01

;< N'15O			
;< P			
			! "# $%
			&' () *+,
			&'
			!
			! -
			! . /0
			! - 1 2 3$
;<	QR		
			45 6 (7

;< S%-T 1 QR			
			8 9 : ;2 <=>? "
;< QR			
			@ A B >C !
;< UV W G			
			D ? E F GH
			D ? I J
			D ? KL D GH
		!	=M N O GH
"		! "	=M N O GH
		!	=M N O GH
			? $;2 PQ
			? $;2 PQ
			? $;2 PQ
		!	! ? $;2 PQ
"		! "	! " ? $;2 PQ
		!	! ? $;2 PQ

/. 23M CDE F 5> 01

CDE F %X YA			
		!	RS1 T
"			RS1 T 7 U2 V=)
			RS1 T 2W XY> >C Z
#		!	RS1 T GH V=)
CDE F YA			
			RS1 T [\ '
CDE F UV YA			
		!	RS1 T GH]^ _
CDE F %X Z[			
"			RS1 T `" Ta
"			RS1 T Bb,c GH

"		!	RS1 T 2W d7
"			RS1 T 2W e
""			RS1 T 7 U2 f>
$		!	RS1 T GH2 gh*b
CDE F \6 67			
			RS1 T ij k6
			RS1 T ij lm
			RS1 T ij lm
			RS1 T ij lm
		"	RS1 T ij lm
#			RS1 T ij lm
		#	RS1 T ij lm
CDE F] ^_			
			RS1 T Z n[o
CDE F W G `			
"$			RS1 T I J ;p
$			? $;2 PQ
IJK ab			
			RS1 T qr
CDE F = c[YA			
			RS1 T sd tY '
CDE F de f:g			
			RS1 T V=)
$			RS1 T V=)
%			RS1 T V=)
			RS1 T V=)
CDE F h Bi j YA			
"%			RS1 T Pu >v
"		!	RS1 T Pu >v
			RS1 T Pu >v w

/. 23M # DE F 01

# DE F %X YA			
#	!	!	x S1 T
#	!		x S1 T $`
#	!		x S1 T 7 U2 $`
# "	!		x S1 T 7 U2 V=)
# #	!	!	x S1 T GH V=)
#	!		x S1 T y 2 7 U2 V=)
#	!		? $;2 PQ
# DE F %X Z[			
#"	!		x S1 T 2W e
#""	!		x S1 T 7 U2 f>
#"	!		x S1 T ^ 7 U2 f>
#"	!		x S1 T 2W D n[z 0
%	!		x S1 T y 2 7 U2 f>
# DE F YA			
#	!		x S1 T [\ '
#	!		? $;2 PQ
# DE F] ^_			
#	!		x S1 T y 2 Z n[o
IJK ab			
#	!		x S1 T qr
# DE F UV W G 8dg			
#	!		" I J? "i {
#	!		? $;2 PQ
# DE F de f:g			
#	!		x S1 T V=)
# $	!		x S1 T V=)
# %	!		x S1 T V=)
#	!		x S1 T V=)
#	!		x S1 T V=) ? $;2 PQ
# $	!		x S1 T V=) ? $;2 PQ
# %	!		x S1 T V=) ? $;2 PQ
#	!		x S1 T V=) ? $;2 PQ

8d CDE F			
	!	!	^ RS1 T
"	!		^ RS1 T 7 U2 V=)
	!		^ RS1 T 2W XY> >C Z
"	!		^ RS1 T `" Ta
"	!		^ RS1 T Bb,c GH
"	!	!	^ RS1 T 2W d7
"	!		^ RS1 T 2W e
""	!		^ RS1 T 7 U2 f>
	!		^ RS1 T Z n[o
8d CDE F \6 67			
	!		^ RS1 T ij k6
	!		^ RS1 T ij lm
	!		^ RS1 T ij lm
	!		^ RS1 T ij lm
	!	"	^ RS1 T ij lm
#	!		^ RS1 T ij lm
	!	#	^ RS1 T ij lm
8d CDE F W G `			
"$	!		^ RS1 T I J ;p
$	!		? $;2 PQ
8d CDE F h Bi j YA			
"%	!		^ RS1 T Pu >v

/. 23M GH 5> 01

GH A]			
			|}~•
			€•,X
			f „
"			x …
			f = sL*f* † (7
GH %X YA			
		!	J‡

			# ˆa
			J‡ $`
			J‡ 7 U2 $`
			J‡ 7 U2 V=)
			J‡ 2W XY> >C Z
#		!	J‡ GH V=)
		!	‰^ J‡ PQ
			? $;2 PQ
GH %X Z[			
			J‡ `" Ta
			J‡ Bb,c GH
		!	J‡ 2W d7
			J‡ 2W e
			J‡ 7 U2 f>
"			J‡ 2W D n[z 0
"&			J‡ Š ‹ 2W e
GH \6 67			
			J‡ ij k6
			J‡ ij lm
			J‡ ij lm
			J‡ ij lm
		"	J‡ ij lm
#			J‡ ij lm
		#	J‡ ij lm
GH YA			
			J‡ [\ '
			? $;2 PQ
"			J‡ oŒ '
			? $;2 PQ
GH kL[ ]			
			•Y Ž Y o '
			•Y Ž Y o ' PQ
			•Y Ž Y o '
			•Y Ž Y o ' PQ

%		"	•Y Ž Y o '
			•Y Ž Y o ' PQ
&			•Y Ž Y o Z
&			•Y Ž Y o Z
%		#	•Y Ž Y o Z
GH de f:g			
			J‡ V=)
$			J‡ V=)
%			J‡ V=)
			J‡ V=)
			J‡ V=) ? $;2 PQ
$			J‡ V=) ? $;2 PQ
%			J‡ V=) ? $;2 PQ
			J‡ V=) ? $;2 PQ
GH l m n89 o p% Z[			
"			•• ^M‘ `" Ta
"			•• ^M‘ Bb,c GH
"		!	•• ^M‘ d7
""		!	•• ^M‘
GH UV W G			
&			J‡ E F GH
&			J‡ KL D GH
&		!	J‡ =M N O GH
& "		! "	J‡ =M N O GH
&		!	J‡ =M N O GH
&			? $;2 PQ
&			? $;2 PQ
&		!	! ? $;2 PQ
& "		! "	! " ? $;2 PQ
&		!	! ? $;2 PQ
GH UV YA			
		!	J‡ GH]^ _

			J‡ E F GH '
			J‡ KL D GH '
			J‡ =M N O ’2 k6>
"		"	J‡ =M N O ’2 k6>
			J‡ =M N O ’2 k6>
			? $;2 PQ
			? $;2 PQ
			? $;2 PQ
"		"	" ? $;2 PQ
			? $;2 PQ
GH h Bi j YA			
"%			J‡ Pu >v
IJK ab IJK L*			
			“h, "‰ 2Y” k6
			• ‹ “h, "‰ 2Y” .
			Š ‹ “h, "‰ 2Y” .
			– ‹ “h, "‰ 2Y” .
			J‡ qr
IJK L*			
&			“h,
&			“h, Bb,c GH
&			“h, `" Ta
&			“h, `" Ta

/. 23M \6 qr L*

0x30						
...	...	...	...	...	...	...
0x3F						
0x70						
0x71	0					
0x72						

0x73	0					
...	...	...	...	...	...	...
0x7F	0					
0x80						
...	...	...	...	...	...	...
0xBF						
0xC0						
...	...	...	...	...	...	...
0xFF						

/. 23M \6 qr L*

0x30						
...	...	...	...	...	...	...
0x3F						
0x40						
0x41						
0x42						
0x43						
...	...	...	...	...	...	...
0x5F						
0x70						
0x71	0					
0x72						
0x73	0					
...	...	...	...	...	...	...
0x7F	0					
0x80						
...	...	...	...	...	...	...
0xBF						
0xC0						
...	...	...	...	...	...	...
0xFF						
0x200				0	0	0
...	...	...	...	...	...	...
0x2FF				0	0	0

/. 23M \6 qr L*

0x30						
...	...	...	...	...	...	...
0x3F						
0x70						
0x71	0					
0x72						
0x73	0					
...	...	...	...	...	...	...
0x7F	0					
0x80						
...	...	...	...	...	...	...
0xBF						
0xC0						
...	...	...	...	...	...	...
0xFF						
0x200				0	0	0
...	...	...	...	...	...	...
0x2FF				0	0	0

CSR ¯ > -°T •W ± • ' ž • .

! " #$ % ! " &!

Z 45 /65 > ~MC ¡c T zq Œ•j # . ^²{³ > &O 4X ´T œ•qr
„ , }X ,-./ 8 > & µ ± > & ¶ # ´T m‰qr . ¡c ·¸¹T zq , q,
> (_ - ‹ KLM 45 ~ 03 v¶ r . ,-./ N &O >
WPRI(Writes Preserve Values, Reads Ignore Values) s• .

Q b{ #$% |p }. ~ •2 € " •, ƒX „…r †.N ‡ . ˆ € "
‰a E ?ŠW e 9† g K‹m: !BŒ :Z• &|ET Ž ?@ †.1F €
?ŠWr - ‰aN 9 < & SKET ••Wr " EF ‚i ‘ ’†. “ S\ K‹m:
€”N 9 < • –* — ˜U , qrp ™+: Œ 9 <

'() "* %

Z 45 /65 CSR > CD P{ Z & xqOv -°T ' 7 , 9- X Œ• .
^²{³ @ > & pt ´v ºr . »-¼ 65C pt ´ ½¾ , 8 -° (Œ: ¿À)
,-./ pt ´3 N' Á ÂR 6- ‹X š›C :®H• , 45 keC pt ´ H7 C'qO
™ . ,-./ N &O > WLRL(Write/Read Only Legal Values) s• .

45 ,?C —h 4šN 9 < -›. œ - 45 p • - e ‡ ,?C ž
—. SŸ f6 •XET

C WLRL > & -0j- ‹ ´T 6• • , KLM ‘’ ŒQ —˜•Ã " #-v f •
@Är §X :® . »-¼3 Å´ pt - ‹Æ • WLRL > 4T ± Ç P{ …†T f¸W
" # . v f¸ ´X pt - ‹X 65 ´e 65 ~ > ´& k'23 ‰ § ˜¨© .

+,, " '() " %

Z 45 /65 CSR > P{ Z & xqOv ' j-v , B´ 6 §T È •O 4T
± É* pt ´T f¸ . CSR 65& 8 " teC u 7 C' • , 0 N'T º Ê Á 4 O @
´ p-j - $ Ë'3 -0j ´ Ìz Í:Î " # . ,-./ N &O >
WARL(Write Any Values, Reads Legal Values) s• .

WARL > & -0j- ‹ ´T ºR KLM ŒQ —˜•;- ‹ . »-¼ 65´ pt - ‹ÆT ±
WARL > 43• KLM Ï pt ´ f¸W " # . v f¸j pt ´X pt - ‹X
65 ´e { :;<= *+& k'23 ‰ § ˆ¨© .

CSR& x 65C ÂÐ CSR B > &O È j pt ´ EFT ˆÑ• , dR Ÿ' u
ÂÐ CSR q, > Ò pt ´ EF ¦ (Ój- ‹X)´T Ô• Õh . 65 ~
> ´ 65 c&R pt lHR Ÿ 2 . Ô , (CSR& x 65 ke q, > ´ ˆÖ
" # .

. 89 Pi 1 ¡ — “ ¢£ . € : ¤y ˜U — z g :Z € , e fB
€ ," + N 9 < + 1 D g “ ¢£ . € : e fB1F g :Zr ¥¦p j=
† § E¨ € ©K SŸC, qª,- ˜U — zM •

Z CSR > 65 CDW ± 8 CSR > d× (alias)3 ' . @ > H 7 , 65 CDW
± C d×3 ØÙ CSR > H 1 . (CSR& x 65 C 45 ~ &O 65 CD3 ˆÚ• ,
Ò ´X C :®H Ÿ & GH Ô• d× x* 5‰ ´T fÛ .

p CSR´ ˆÚ §X Û¡T ÜX CSR& x 65C :®Ý , q, CSR& • " te
—˜•;- ‹ .

KLM ±± CSRT Þß23 4 . Œ ¶ S àX !"# CSRT Þß23 4 .
dR Ÿ' u , CSR Þß2 45C f¸ ´X áâ "[&O CSR \] q, CSRT
•23 4½T ± f¸j ´e } .

CSR ã ˆÚ• , Œ ¶ ” mstatus ,-./ 5Ê ISA (–& N x SXLEN “ UXLEN ˆÚ• ,
dR Ÿ' u Ò ãT Õ CSR 65 CD > ž P{ ´X ~ ã CSR / • Í7¿äT
2 §=å k' .

1. ~ ã CSR ´T }X ã Ç• ,-./& æ .
2. ~ ã CSR&O 45 ~ P{ž }X zb& # Ç• ,-./ P{ 03 N' .

3. Ç• ,-./ ãT Ò ã3 ˆç . Ò ã WC ~ ãm °3• Ç• ,-./ z WP{
p- 7 *z P{ ¤è . Ò ã l é• Ò ã& êh 0 $% .

4. Ò ã CSR& # ë 65 CD > Ç• ,-./ }X zb P{ ´T Õ .

CSR ãT ˆÑ §X q, CSR 45 65C :®Ý " te —˜•;- ‹ .

s[CSR XLENP{m ì3• , •2 45 ,-./ z XLENP{ f¸ 7 •2 65 z XLEN
P{v '" Y *z P{ @x p- .

Zicntr $% «¬/ CSRe }X Z s[CSRX XLEN=32(RV32)Z ±R É* 64P{ . @ ë 64P{
CSR(Œ: «¬/ "$%&)& xy }X í î& ï1 ' ð 32P{ *z Žf CSR("$%&') ' . *z Žf
CSRX xñ 64P{ CSR P{ 63:32& x d× . GHO RV32 ^²{³ 0Ä \]W " u *z
32P{ 4¾ '"W " # .

s[*z Žf CSRX 5Ê RISC-V EF RV32(XLEN=32)Z ±v \]W " # . RV64(XLEN=64)&O
s[*z Žf CSR ^C Œ•j #3Ý , CSR& \] • • Zf23 ‘’
ŒQC —˜ .

-. / 01

dT#• $ %&' % ()% * dTs +' $ % & ,-()% '& %, * |— ˜Y q™š ›œz •ž Ÿq? ‘
¡ ¢ $ %&' % ()% * ? |— ¡ €N£ •E 2 ¤¥ Ž¦§ |¨ © š ª£

. $ %&' % ()% * dT#•« ' (/'))% #0 12,()% %&' % ()% * 3 ¬b |z£ ‘ ¡- ™— ®ƒ=¯
Q²£ dT#•? o] 7 ° k †±« €+ O²³ ´µ£

d – 4 T# . 5 6 #%7 '8 &' %9 ,% ,: "9 1 #/1;1'%1 9 /17 < 0')%9 = &' # %)/1> 3

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